

SolaForm Artificial Leaf

Solar-Powered Carbon Capture and Formate Production

Technomania / RoboRumble STEM Innovation • 2026

Team: SolaForm Artificial Leaf

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Executive Summary

SolaForm Artificial Leaf is a proposed standalone, solar-powered carbon-management system designed to capture carbon dioxide (CO₂) from ambient air and convert captured carbon into formate while producing oxygen. The concept integrates an air-moving fan, an amine-based CO₂ sorbent, solar-assisted thermal regeneration, a CO₂ buffer, water storage, an electrochemical conversion stage and an embedded control system. The immediate competition prototype is software-first and demonstrated through Wokwi. The simulation focuses on control logic, sensor interpretation, state transitions, actuator control, product accumulation and safety behaviour, rather than claiming laboratory-validated chemical performance.

1. Title Header

Team	SolaForm Artificial Leaf
Build / Solution	SolaForm Artificial Leaf – Solar-Powered Carbon Capture and Formate Production
Competition Category	Technomania / RoboRumble STEM Innovation
Prototype Stage	Wokwi software simulation; physical prototype proposed for later validation

2. Context & Problem Statement

2.1 Background

Carbon dioxide emissions are associated with electricity generation, transport, industrial activity, fuel use and other energy-intensive processes. The project identifies a need for carbon-management approaches that can be modular, renewable-energy powered and suitable for distributed operation. The supplied project presentation further identifies industrial emissions, transportation fossil-fuel combustion and the burning of coal, oil and gas as contributors to increasing atmospheric CO₂.

2.2 Problem

The core problem addressed by SolaForm is the continued accumulation of atmospheric CO₂ and the need for practical systems that can capture CO₂ and convert it into a potentially useful product. Existing carbon-management approaches can be centralised, energy intensive or difficult to deploy at small scale. In addition, solar availability varies, creating a need for intelligent automated control of energy-demanding processes.

2.3 Problem Validation

The project materials validate the problem primarily through background research and problem analysis. The presentation links industrial emissions, vehicle fuel combustion and fossil-fuel use to increased atmospheric CO₂ and identifies environmental and economic impacts including global warming, extreme weather, rising sea levels, biodiversity loss, infrastructure damage, agricultural impacts and increased health-related costs. The business

proposal recommends further customer validation through controlled pilots, measurable performance data and early engagement with industrial, research and sustainability organisations.

3. Solution Overview

SolaForm is a modular solar-powered system that combines CO₂ capture, regeneration and electrochemical conversion under automatic control. A fan draws ambient air through an amine-based sorbent. When the capture stage reaches a defined condition, the controller initiates regeneration using a solar-assisted heater. The released CO₂ is directed into a buffer and then supplied with water to a simulated electrochemical conversion stage. The intended outputs are formate and oxygen. A product tank tracks formate accumulation, and conversion pauses when the tank reaches capacity.

3.1 Operating Cycle

Stage	Process	Control / Output
1. Solar check	Check simulated solar availability.	Operation proceeds only when the required resource is available.
2. CO ₂ capture	Fan moves ambient air through sorbent.	Simulated CO ₂ concentration tracks capture progress.
3. Saturation decision	Determine whether capture threshold has been reached.	Controller switches to regeneration.
4. Regeneration	Solar-assisted heater releases captured CO ₂ .	Concentrated CO ₂ moves to buffer.
5. Conversion	CO ₂ and water enter the simulated electrochemical stage.	Formate and oxygen outputs are simulated.
6. Product accumulation	Formate is accumulated in the product tank.	Tank level is monitored.
7. Tank-full control	Conversion stops at capacity.	Warning displayed; unsafe progression prevented.
8. Operator interaction	Button simulates emptying the product tank.	System can resume after collection.

4. BOM Summary

The project materials identify the following core hardware/software elements. The current prototype is a Wokwi simulation; the physical chemical components are proposed for later validation. The competition-specific Excel BOM was not included among the supplied documents, so clickable links to the main Excel BOM cannot be inserted without that file.

Core component	Purpose
Arduino-compatible controller	Runs the state machine and interprets sensor inputs.
Fan / air-handling component	Moves ambient air through the CO ₂ sorbent.

CO ₂ sensor / simulated CO ₂ input	Represents CO ₂ concentration and capture progress.
Temperature sensor / simulated temperature input	Monitors regeneration conditions.
Water-level sensor / simulated water input	Prevents conversion when water availability is insufficient.
Solar input / simulated solar availability	Determines whether solar-powered operation can proceed.
Heater / simulated heater	Represents thermal sorbent regeneration.
Servo / flow-control actuator	Demonstrates controlled movement/flow switching in the simulation.
OLED display	Displays system state and simulated formate/oxygen production.
Product tank variable / storage	Tracks accumulated formate and tank-full condition.
Water storage	Provides the water input for the proposed electrochemical stage.
Amine-based sorbent	Captures CO ₂ ; physical performance remains to be validated.
Electrochemical conversion cell	Proposed stage for converting CO ₂ and water into formate and oxygen.
Solar panel and power electronics	Planned physical energy source for later prototype validation.

5. Marketability & Validation

5.1 Target Users and Market

The recommended market-entry strategy is business-to-business and research-to-market rather than immediate mass consumer sales. Potential customer groups include industrial and commercial facilities seeking distributed carbon-management solutions, cleantech and renewable-energy organisations, universities and research institutions, sustainability-focused organisations, and remote/off-grid sites where autonomous solar operation may be valuable. The project presentation additionally identifies schools and universities, research institutions and environmental organisations as target markets.

5.2 Value Proposition

- Combines CO₂ capture, regeneration and conversion in one modular concept.
- Uses solar energy as the proposed renewable energy source.
- Uses embedded control to automate the operating sequence and reduce continuous supervision.
- Aims to convert captured CO₂ into storable formate while producing oxygen.
- Uses a software-first prototype to test control logic before committing to more expensive physical development.

5.3 Marketing Strategy

- Social media and digital technical demonstrations to communicate the concept and prototype.
- School and university partnerships for STEM demonstrations, research and early user feedback.
- Science fairs, innovation competitions, research conferences and exhibitions.
- Environmental awareness campaigns explaining carbon capture and carbon conversion.
- Strategic partnerships with laboratories, engineering organisations, solar suppliers and cleantech incubators.
- Pilot agreements with suitable industrial or commercial partners once physical validation is achieved.

5.4 Customer Feedback and Validation Structure

Customer feedback should be collected progressively rather than relying only on the competition prototype. The proposed structure is: (1) demonstration feedback from students, researchers and STEM audiences; (2) structured interviews or surveys with potential institutional and industrial users; (3) controlled pilot testing with measurable performance indicators; and (4) refinement of the product, controls and business model using pilot results. The supplied proposal recommends early customer validation and treating formate sales as a hypothesis until product quality, production cost, market demand and regulatory requirements are established.

5.5 Revenue Model

- Paid feasibility studies and pilot installations.
- Demonstration units and engineering/design services.
- Sale or lease of validated complete units.
- Monitoring software and service contracts.
- Replacement sorbent/consumables and maintenance.
- Technology licensing.
- Potential formate sales only after technical, economic and regulatory validation.

6. Technical Implementation

Technical implementation is presented at the lighter level appropriate to Technomania. The focus is on feasibility, prototype functionality and a clear development pathway rather than extreme optimisation.

6.1 Mechanical Implementation

The proposed mechanical arrangement consists of an air inlet and fan, a CO₂-capture sorbent section, a regeneration/heating section, a CO₂ buffer chamber, water storage, an electrochemical conversion stage and product storage. The system is intended to operate as a modular unit. In the Wokwi stage, physical airflow, sorbent chemistry and electrochemical reactions are represented by software-controlled inputs, actuators and variables. The physical mechanical design remains a later engineering-validation task.

- Air handling: fan represents movement of ambient air through the sorbent.
- Regeneration: heater represents solar-assisted thermal release of captured CO₂.
- Buffering: a dedicated buffer represents temporary storage of concentrated CO₂.
- Water management: water storage supplies the proposed conversion stage and is monitored for availability.
- Product storage: an internal product tank represents formate accumulation and tank-full control.
- Safety: physical containment and engineered safeguards will be required before chemical operation.

6.2 Electronic Implementation

The electronic/control architecture uses an Arduino-based controller in the Wokwi demonstration. Simulated inputs represent solar availability, CO₂ concentration, regeneration temperature and water level. Outputs include fan/heater representations, servo movement for flow control, an OLED display and product tank logic. The controller continuously interprets system conditions and selects the appropriate operating state.

Electronic element	Role
Controller	Runs automatic state machine
CO ₂ input	Represents capture progress
Temperature input	Represents regeneration condition
Water-level input	Represents available water
Solar input	Represents available solar power
Fan/heater outputs	Represent air movement and regeneration
Servo	Demonstrates flow-control movement
OLED	Displays state and product information
Button	Represents product-tank emptying

6.3 Programming Implementation

The programming approach is a state-machine control system. Sensor values are interpreted against defined thresholds and the controller transitions between solar checking, capture, regeneration, conversion and tank-full states. The program also manages simulated actuator behaviour, OLED messages, product accumulation and fault/safety conditions. The Wokwi implementation validates software workflow but does not validate actual chemical capture or conversion performance.

Core programming logic:

- Check whether solar availability is sufficient.

- Run the capture state while the simulated CO₂/sorbent condition permits.
- Transition to regeneration when the capture threshold is reached.
- Regenerate and route concentrated CO₂ to the buffer.
- Check water availability before conversion.
- Run simulated conversion and increase product-tank level.
- Pause conversion and issue an alert when the tank is full.
- Allow operator button input to simulate product collection and tank reset.

7. Prototype Testing & Validation

The supplied proposal defines validation around measurable control, energy, capture, regeneration, conversion, product-quality, autonomy and safety indicators. At the current software stage, the strongest validation evidence is control-system behaviour in Wokwi. Physical chemical performance must be tested later.

KPI	Planned measurement
Control reliability	Correct transitions between capture, regeneration and conversion under changing simulated inputs.
Solar utilisation	Share of operating demand supplied by available solar energy; minimise unnecessary fan/heater operation.
CO ₂ capture performance	CO ₂ captured per cycle and per unit of sorbent during physical testing.
Regeneration performance	CO ₂ released and energy required per unit of CO ₂ released.
Conversion performance	Formate yield/selectivity and oxygen generation during physical testing.
Energy efficiency	Energy consumed per unit of formate and per unit of CO ₂ processed.
Product quality	Formate concentration and relevant purity specifications.
Autonomy	Number of cycles completed without human intervention, excluding product collection.
Safety	Response to low solar power, high temperature, low water and full product-tank conditions.

8. Development Roadmap

- Phase 1 – Simulation: complete Wokwi state machine, thresholds, actuator logic, OLED interface, product accumulation and fault handling.
- Phase 2 – Bench control prototype: build low-voltage electronics and verify sensors/actuators independently of chemistry.
- Phase 3 – Component validation: characterise airflow, sorbent capture/regeneration and solar-heater performance.

- Phase 4 – Electrochemical validation: test-controlled CO₂ and water inputs and measure formate and oxygen outputs.
- Phase 5 – Integrated prototype: combine validated subsystems, safety interlocks, data logging and automatic control.
- Phase 6 – Pilot: deploy with a suitable partner and collect performance and cost data.
- Phase 7 – Commercialisation: standardise manufacturing, compliance, service model, pricing and sales partnerships.

9. Risks, Assumptions & Mitigation

Risk	Potential issue	Mitigation
Chemical performance	Conversion may not achieve assumed yield/selectivity at acceptable energy cost.	Laboratory validation before commercial claims.
Energy balance	Solar energy may be insufficient for fan, regeneration and conversion.	Energy budgeting, duty-cycle optimisation and storage where justified.
Sorbent durability	Capture capacity may decline over repeated cycles.	Cycle-life testing and replacement strategy.
Product market	Formate may not have sufficient value to support economics.	Validate customers and product specifications early; treat sales as a hypothesis.
Safety	CO ₂ , heat, pressurised components and electrochemical equipment create hazards.	Containment, sensors, interlocks and professional safety review.
Prototype-to-product gap	Successful Wokwi simulation does not prove chemical or commercial performance.	Separate software, engineering and laboratory validation gates.

10. Team Roles & Contributions

Team member	Role	Key responsibilities
Ketsephile Rossa Ratshilaya	Research and Documentation	Research on artificial photosynthesis, direct air capture, carbon conversion and formate production; concept validation; Folder C documentation; marketability/commercialisation strategy; cost analysis and supplier sourcing.
Refiloe Primrose Masebe	Systems Development & Software Lead	Refined the concept into an implementable solution; defined operating stages and workflow; Folder A source

		code; Wokwi simulation and software implementation; programming/control documentation; simulation testing.
Nathan Maleka Mashego	Concept Developer & Design Lead	Initiated the SolaForm Artificial Leaf concept; identified rising atmospheric CO ₂ as the environmental challenge; proposed the artificial-photosynthesis approach; Folder B documentation; simulation design and testing.

11. Evidence, Supporting Pictures & Screenshots

For the final Technomania submission, this section should contain visual evidence from the project. The supplied documents confirm that the design and Wokwi simulation were developed, but the actual screenshots/images are not embedded in the parsed source material available for this report. Insert the following visuals from the team's original files:

- Wokwi circuit/simulation screenshot showing the controller, sensors and actuators.
- Wokwi control-logic/state-machine screenshot or flow diagram.
- OLED output showing operating state and simulated product information.
- Screenshot of the tank-full warning and operator reset interaction.
- Mechanical/system architecture diagram showing air → sorbent → regeneration → CO₂ buffer → conversion → product storage.
- Relevant source-code screenshots demonstrating state transitions and safety logic.
- Physical prototype photographs when the hardware validation phase begins.

12. Technomania Marketability Summary

SolaForm is positioned as a modular carbon-management platform rather than only a chemical device. Its early commercial strategy is to demonstrate reliable automatic control, validate the underlying chemistry and energy balance, and then conduct sponsored or paid pilots. Potential commercial offerings include modular units, leasing, monitoring, maintenance, consumables, engineering services and licensing. The project deliberately avoids treating simulated chemical outputs as established results; formate yield, oxygen generation, product quality, energy efficiency and commercial economics remain validation targets.

13. Conclusion

SolaForm Artificial Leaf demonstrates a software-first pathway toward a solar-powered system for CO₂ capture, regeneration and conversion. The Wokwi prototype provides a practical proof of concept for automatic process control, sensor interpretation, state

transitions, actuator control, product accumulation and safety responses. The proposed physical system would extend this work by validating sorbent performance, regeneration energy, electrochemical formate production, oxygen generation, product quality and overall energy-economic feasibility. The project's strongest route to market is controlled validation followed by pilot deployment with suitable research, sustainability, cleantech or industrial partners.

14. References

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